

Party planning & booking system

CS 1151_Fundamentals of Object Oriented Programming project

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Features:



Description:

A system used by party planners to make client booking more convenient, displaying all services offered transparently by displaying the service details, duration, and prices. It helps party planners keep track of all their clients, their needs, and any specific details they require.

Party type

Birthday, Baby shower, Graduation

Venue

Home, Garden, Ballroom/hall

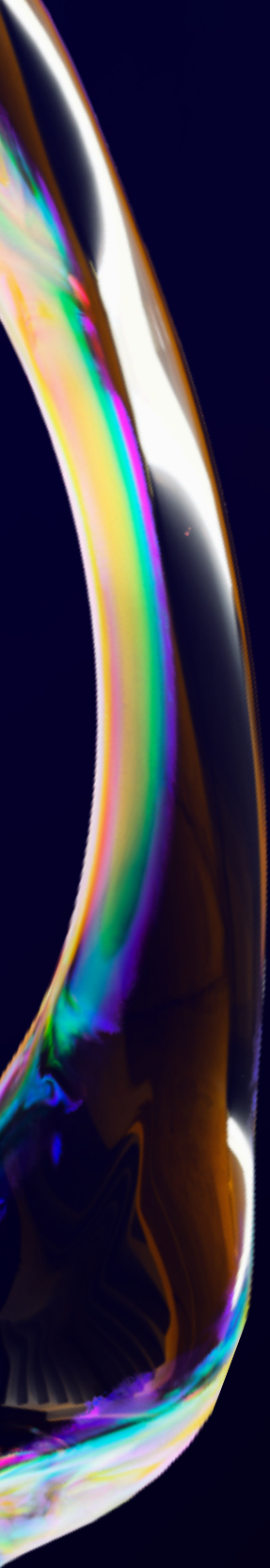
Services

Decoration packages (Standard, Luxury, Premium) & Catering options (Basic, Premium, VIP). Photography, Videography, Music, Entertainment & Kids Activities.

Booking

Based on availability. The system will use user preferences and output a final bill and detailed plan for the party which includes all client specifications.





OOP Principle	Implementation
Encapsulation	All attributes are private/protected with public getters/setters
Inheritance	Client inherits from Person; Catering/Decoration/Entertainment/Photography inherit from Service
Polymorphism	Virtual functions displayInfo() and displayService() with overrides in each subclass
Composition	Booking class contains Client, Event, Venue, and Service objects (HAS-A relationships)
Constructors & Destructors	Constructors were used to initialize objects and set default values, while destructors were implemented for proper object cleanup
Exception Handling	Exception handling and input validation were used to detect invalid user inputs and improve program reliability.
Memory Management	Proper use of constructors/destructors with new/delete for dynamic allocation

Class & Responsibility

Class Client

This class responsibility is related to the users specific data such as payment methods booking history. Service providers

Class Event

This class coordinates the details , its job is to link a specific client to a specific venue and list the services.

Class Venue

This class that specifies the physical location where the event happens it stores details about the venue. Such as (location, capacity , rental price ,indoor or outdoor)it also checks the availability of the venue at that specific date.

Class Booking

This class connects all classes together. Stores client and event information. Calculates total price. Handles selected services. Displays final booking summary.



Class & Responsibility

Class Service

it is a base class of all services the company provides it holds the basic items that every class has (Name, price).

Class Catering

it is a class that inherits from the base class (service) which stores anything related to food and beverages that the company can provide. It includes a menu and guest count .

Class Decoration

It inherits from the base class (service), manages the theme of the event .

Class Entertainment

It inherits from the base class (service), handles the scheduling and requirements for performers, DJs, or guest speakers.

Class Photography

It inherits from the base class (service), manages the media requirement , number of photographers ,hours of coverage.